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Weathering steel – 30 years of experience in Banedanmark

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ABSTRACT

Banedanmark has used weathering steel for more than 30 years and has gathered unique information about corrosion of weathering steel during this period. Not only has the corrosion rates been noted - but also the good and poor detailing. The conclusions of this will be given in this paper. Based on the information gathered from the tests, a method for estimating the corrosion lifetime for the specific type of structure is given.

Since 1986 weathering steel has been widely used on the Danish rail network for overhead line equipment structures such as masts and frames etc. As these structures now represent a very large investment to the owner of the rail net, it is crucial to follow if the expected lifetime from design is valid.

These structures are based on a 'standard-design' developed by Rambøll in the mid 80ti'es. At that time the knowledge on how weathering steel would behave on these specific structures was limited, hence a test-programme was carried out from 1988 to 1994, in order to verify the corrosion rate in general, along with the suitability of the design philosophy of details. This test-programme was carried out using designated test-masts that was located along the rail track at selected locations.

After seven years of studying the test-masts, the test-programme was stopped, as these results indicated that the required service life would be met. And there was not observed any critical items on the detailing used.

Now, after almost 30 years of service, the designated test-masts are still available, and as results from the first test-regime indicated some conservatism in design assumptions, it might be the case that the existing structures can be considered safe beyond the original design life of 50 years.

To investigate this further a pilot test-has been carried out, to check if the test-regime is still suitable for gathering new information after almost 30 years of exposure. This initial test showed that it might be possible to increase the expected service life from 50 to perhaps 80-100 years of existing structures.

In addition it was evident that the design philosophies of details are suitable for long time exposure. It even indicated some conservatism in the philosophy of detailing too.

The tests have gathered unique knowledge of how mast structures for overhead lines of weathering steel behaves after long time exposure.

Keywords: Weathering steel, COR-TEN steel, Electrification projects

1 INTRODUCTION

1.1 Structures for overhead line equipment in Denmark

In 1979 the Danish Parliament decided to initiate electrification of the main railway lines by use of overhead line equipment (OLE). This included installation of an overhead catenary system, bridge alterations, electricity supply etc. As 2100 km of single track was involved, it was the largest investment in a high technology project by Banedanmark (the Danish State Railways (DSB) at that time) [1].

A thorough development phase took place in the years 1979 to 1985 including initial installations. The philosophy was to create a 'year 2000 design' implying that it should be a modern design with the newest techniques.

Great attention was put to the supporting masts of the catenary system. Both in terms of function, economy, maintenance and service but also on the aesthetics as 40,000 masts was expected it would impact the landscape of the country.

One of the design requirements was to have the structures free from regular maintenance for the first minimum 50 years. This could only be met by concrete masts as aluminium and stainless steel was not economically competitive and any surface treatment of regular steel could not guarantee the full 50 years.

After concrete masts were installed on a test-section of the railway track, the design was changed to comprise of masts in weathering steel. This was an innovative approach as they were economical attractive due to the lack of surface treatment, they fulfilled the demand for maintenance as there was none and it leaved a discreet natural imprint on the surrounding landscape.

This innovative design of steel structures of weathering steel that turned out to be strongly competitive to concrete options was then awarded with the European Award for Steel Structures in 1987 (the ECCS Design Award).

The concept of using masts of weathering steel has been used ever since on the main railway lines in Denmark. In fact it is basically the same design that is used in recent years, where additional electrification projects is carried out, but with some modifications as the electrical equipment and other design requirements has been updated since the 1980s.

An example of the structure can be seen in *Fig. 1* where it can be seen that the mast comprises of rectangular or square hollow sections (SHS) and in case a horizontal cross bar is required it is typically a circular hollow section (CHS). In areas where several railway tracks are spaced closely together, similar supporting structures are made as slender frames of SHS in weathering steel.

1.2 Design philosophies in original project

Structures of weathering steel rely on the formation of a dense and well-adhering layer of corrosion products. When this layer is formed, the corrosion rate of weathering steels is significantly slower than of regular mild steels. This layer has the characteristic dark brown rust colour of weathering steel.

The formation of the corrosion layer usually requires regular dry/wet cycles in a non-aggressive environment. If the structure is constant dry (i.e. indoor) the weathering steel will look as regular mild steel without the formation of the rust layer, and if constant wet the weathering steel will corrode fast as regular mild steel.

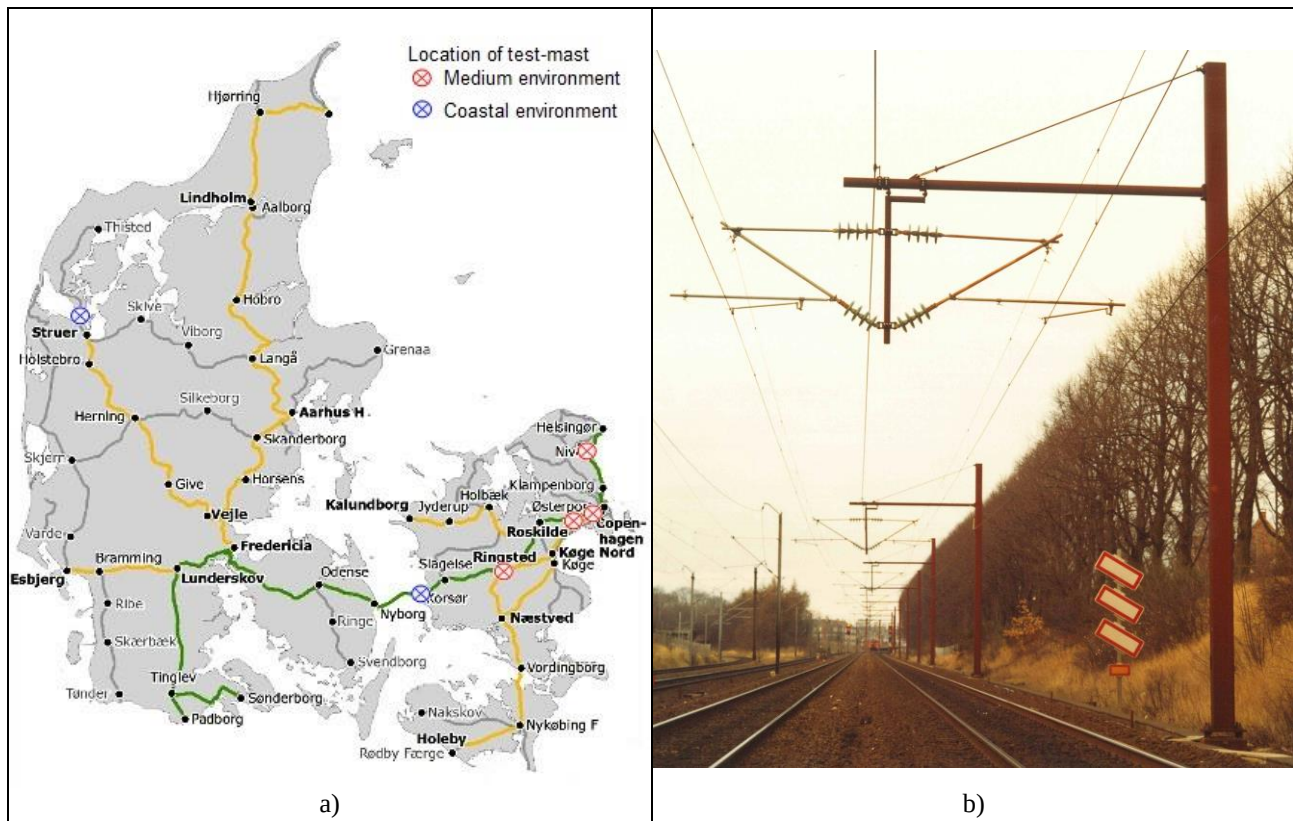


Fig. 1. a) selected locations for test-masts b) mast made from weathering steel. Vertical SHS and horizontal CHS.

Thus designing structures of weathering steel requires specific attention on the detailing, as an inappropriate detail might create a risk of accelerated corrosion. I.e. if a poor detail results in ponding water or collection of corrosion products, which will increase the time of wetness on the local spot by a magnitude.

As 50 years of service without any maintenance was required to the primary structures, the following design philosophies was introduced in the project. The primary intension is to secure that all faces of the weathering steel that are exposed to atmosphere is well ventilated, and that risk of ponding water or collection of corrosion products is prevented.

Design of joints:

- Thin plates shall have a gab at least 3mm
- Thicker plates can be allowed contact under certain conditions
- Clamps shall have a gab of at least 3mm to the structure
- All welds shall be continuous and fully closed
- Confined spaces shall either be closed airtight or secured to be well ventilated
- Fasteners shall be stainless A4
- Washers are used under both head and nut of bolts

The following shall be avoided:

- Details that can collect dirt and corrosion products
- Ponding water
- Narrow cavities
- A gab of less than 3mm to any other structure
- Aluminium alloy in direct contact or below the weathering steel
- Structure below the weathering steel where a discolouring from corrosion products cannot be accepted

Thick plates may be allowed direct contact in a connection given that the connection is detailed such that the contact pressure is large enough and that bolts are placed closely together making the connection sufficiently rigid.

Many various components of the electrical equipment should be fastened to the masts. To have a flexible design, there was a great interest in using clamp fittings. However clamps are critical, as these would not follow the design rules of well-ventilated faces and clamping on a CHS or SHS would not provide sufficient contact pressure on the entire interface to keep water out. Consequently a large effort was put into designing the details of the clamps to allow perfect corrosion conditions of the weathering steel.

Only bolts, nuts and washers of stainless steel were used even it might introduce some increase in the corrosion rate locally at the edge of the washer.

Corrosion products of the weathering steel will be washed off by rain, resulting in leaching of cobber. This could increase the corrosion of any less noble material (such as aluminium alloys) it passes and the magnitude of this effect was not studied at that time. In addition the corrosion products will discolour any structure below.

A selected few of the typical details that are used throughout the project and that follows the design philosophies above is shown in *Fig.2* which is taken from [2].

The structures were in general designed to have a sacrificial thickness of weathering steel of minimum 0.6mm of every exposed face, which was assumed to corrode during 50 years of service [3].

1.3 The test-programme on corrosion

When the original project was designed, knowledge of the general corrosion of weathering steel was available, but not at specific structures comparable to the masts supporting overhead line equipment. The experience in using weathering steels was somehow limited, and there had been a few poor examples without a design philosophy for detailing leading to accelerated corrosion of weathering steel. This lead to the publishing of the fact sheet [4], that contains guidance on the detailing and expected corrosion rates in Denmark. The design of the masts was compliant to these guidelines.

In order to gain further knowledge on how weathering steel performed on the specific structure, and to secure that the very large investment was made on the right basis, a separate test-regime was prepared in order to give an early guidance if the 50 years of maintenance free period was realistic or not.

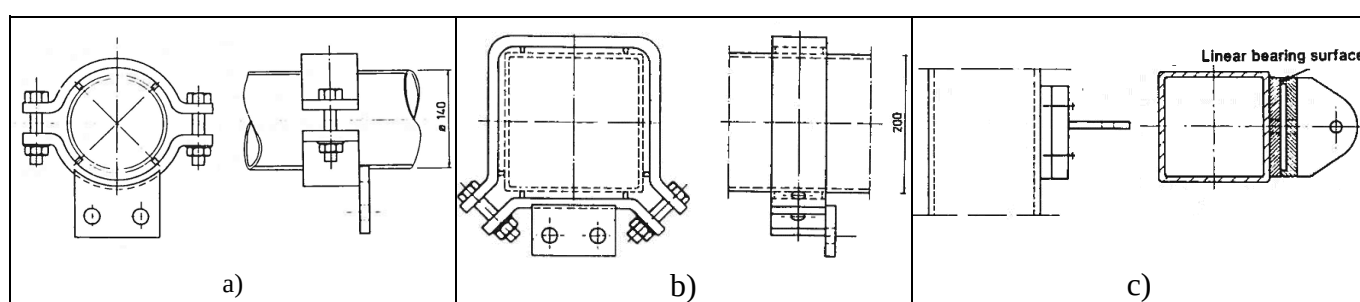


Fig.2. Typical details from the original project with min 3 mm spacing
a) clamp on CHS; b) clamp on SHS; c) accommodation fitting

This test-programme had two main objectives [3]:

1. *To determine the corrosion rate of weathering steel on different locations in Denmark*
2. *To investigate corrosion conditions on details that was not declared safe in advance for long term exposure*

The test-programme was designed and planned to run for 17 years. Hence it was assumed that before the end of this, it would be possible to determine if the 50 years was a realistic requirement based on the obtained data.

The test-regime consisted of designated test-masts on selected locations, which can be seen in *Fig. 1*. The locations were selected based on the expected environment such that areas of coastal, industrial, urban and rural environment were represented. The designated test-masts were applied with various types of dummy-details that were not part of the actual line equipment, but only served the purpose of this test-programme. The outline of the test-masts can be seen in *Fig. 3*.

In order to determine the corrosion rate, each of the test-masts was equipped with plates that only had a single face exposed. The backside and the edges were protected from atmospheric exposure leaving only the front exposed for corrosion. At each mast these plates were labelled as

- *Type A: The exposed face is vertical*
- *Type B: The exposed face is horizontal facing upwards*
- *Type C: The exposed face is horizontal facing downwards*

By doing so it has been possible to distinguish corrosion rates further into direction of exposure.

Each year a selected number of test-plates type A, B and C were to be gathered from the test-masts, and in a laboratory the amount of corrosion was measured. This was done by gently cleaning the plates using solvents only as any mechanical cleaning would disturb the results. Then the plates was immersed in hydrochloric acid added an inhibitor in order to remove the dense layer of corrosion products. Finally the rate of corrosion was calculated based on the loss of weight of steel during the period of exposure.

All plates were remounted at the same location and orientation on the test-masts after determination of weight loss, but as the dense layer of corrosion products was removed in the laboratory, the time of exposure of each plate was restarted. However as not all plates was collected each year but a planned schedule was followed, results was obtained for various lengths of exposure.

In order to determine corrosion conditions on details, each of the test-masts was equipped with various details. Both details similar to the ones used on the actual masts and that followed the previous mentioned design philosophies, but also details that intentionally did not follow the design philosophy. I.e. the test-mast consists of clamps both with and without the 3mm gaps shown in *Fig.2* and with plates joined directly face-to-face without the 3mm spacer. All bolts on the test-masts were stainless A4, hence the risk of increased corrosion could be investigated also and stainless bolts were even mounted into threaded holes cut into the weathering steel.

The original electrification project was made of COR-TEN steel, which is a brand of weathering steel belonging to U.S. Steel. The test-plates and details on the test-mast was therefore also made from COR-TEN steel, but a local Danish supplier, Det Danske Stålvalseværk (DDS) had a second set of test-plates types A, B and C mounted too. Thus it was possible to distinguish results between two different suppliers of weathering steel also when considering the corrosion rates.

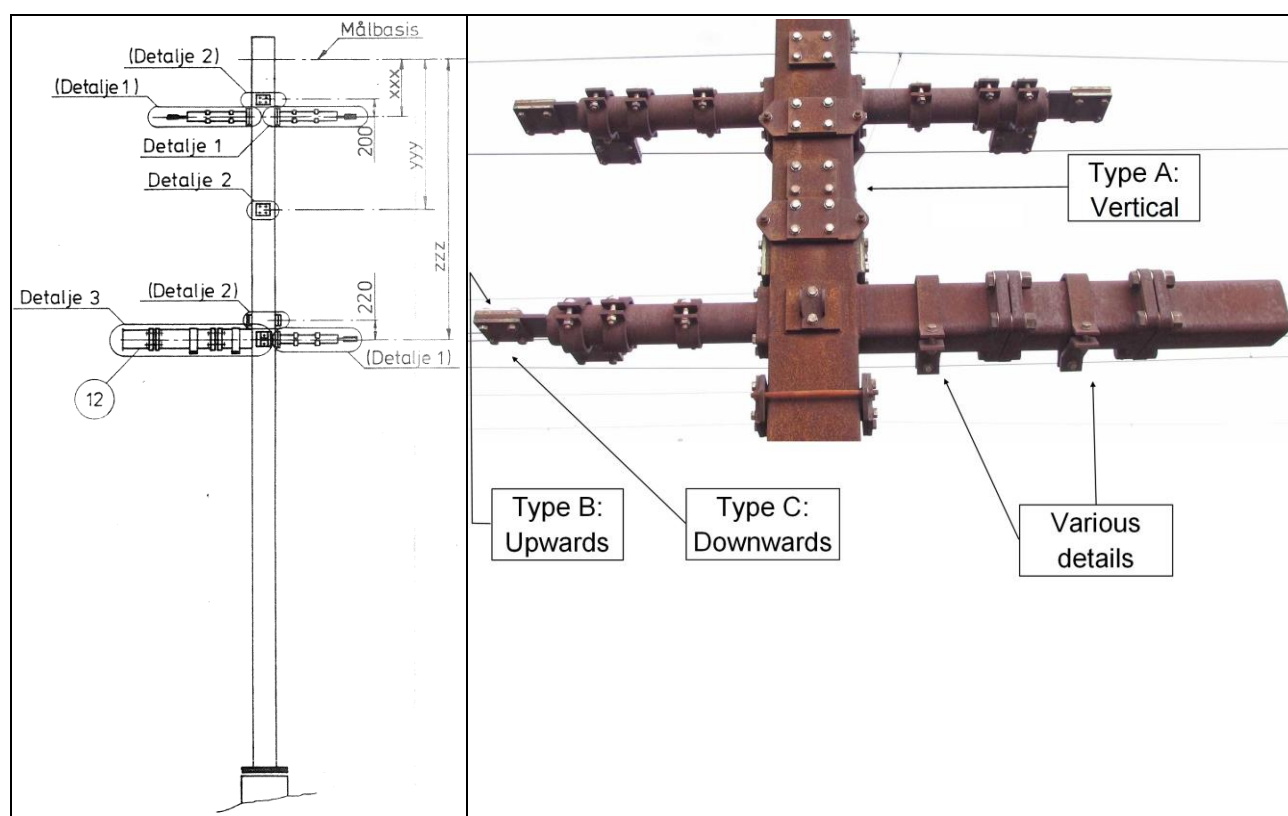


Fig.3. The designated test-masts

The original test-programme was commenced in 1988 and terminated after 7 years in 1995 leaving results from 1 to 7 years of exposure. However the designated test-mast including their special test-plates and details remained in operation.

2 PREVIOUS CONCLUSIONS

The original test-regime was ended after 7 years of exposure. At that time the following conclusions were drawn;

- The sacrificial thickness of 0.6mm appeared to be safe for 50 years of exposure
- The details that followed the design philosophies appeared to be safe for 50 years of exposure
- The use of stainless bolts appeared to be safe for 50 years of exposure
- The corrosion rate of COR-TEN and weathering steel from DDS was similar
- There was a clear difference in the corrosion rate of plate type A, B and C depending on the orientation of the exposed face. The type C with exposed face downwards was corroding significantly faster at every single location
- The deviation between different environments was not clear. It was concluded that the locations could be divided into two groups; coastal environment and medium environment
- The corrosion of test-plates type B in a medium environment can be predicted by $C = 50 \cdot t^{1/3}$
- The corrosion of test-plates type C in a medium environment can be predicted by $C = 60 \cdot t^{1/2}$

The deviation on locations into coastal and medium environment is shown on Fig. 1. These terms are used throughout this paper.

The entire population of data that was collected during the first 7 years is shown in Fig. 4. The rates of corrosion are given depending on the environment and direction of exposure.

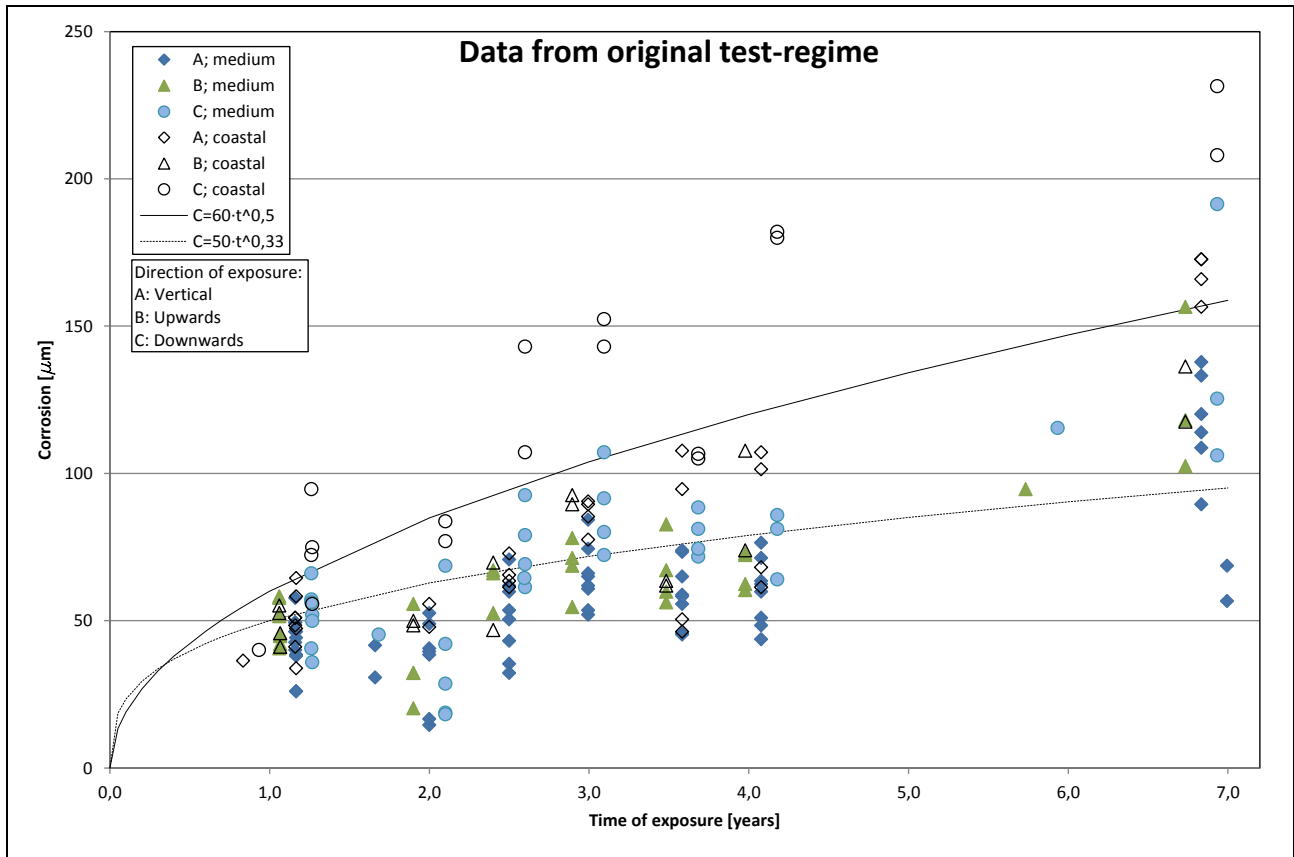


Fig.4. Data of atmospheric corrosion from the previous test-regime

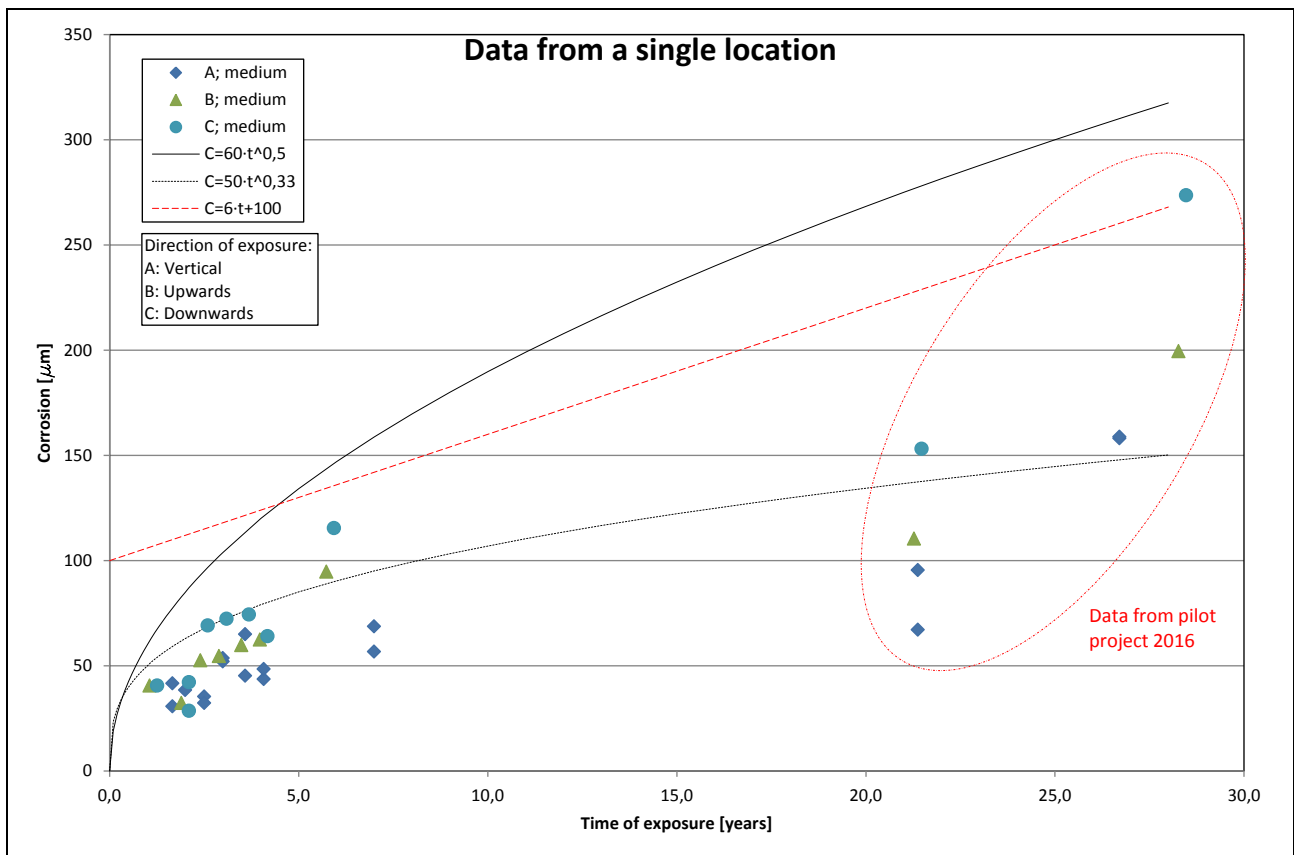


Fig.5. Data of atmospheric corrosion from test-mast of the pilot test

3 RESUMPTION OF TEST-REGIME

Originally the test-regime was meant to confirm if assumptions in the design was appropriate. But as the test-masts are still in operation and as the entire amount of structures represent a very large investment, it has been considered in year 2016 whether the test-regime can be continued. Future results could potentially give an indication if the expected service life of 50 years can be decreased or increased. And as it can be seen from the previous conclusions, it was indicated that less than 0.6 mm corrosion would be expected after 50 years of exposure.

In 2016 a pilot project was carried out in order to determine if a resumption of the test-regime was realistic by having one of the test-masts was disassembled and investigated. The corrosion rates were determined on the plates type A, B and C and the details was investigated carefully. The test-mast for this pilot test-was located at a mild environment.

4 30 YEARS CORROSION RATES

The corrosion rate determine in the pilot project is shown in *Fig 5* together with the corrosion rates at the exact same location from the original 7 years testing. A few of the plates had never been collected for determination of corrosion rate, thus they had the full 28 years of exposure. The other plates had been used before after a few years, hence the time of exposure had been restarted on these.

After long term exposure the previous conclusions seem to be valid still, including the pattern of type C being the plate that corrodes faster.

5 SUITEABLE DETAILING

When considering the details of the test-mast little difference was noticed in corrosion of details that follows the design philosophy and details that does not. Even the clamps that were made of a poor design by intension did not show significant more corrosion than clamps with spacers. Thus it appears that the design philosophy is too conservative for detailing in a mild environment. Refer *Fig 6* that shows clamps with and without spacers.

It should however be studied further on more locations before a clear conclusion can be drawn. But nothing was observed that indicated that the design philosophies used would not give 50 years of service life.

In addition to the design of details, it has also been noticed that graffiti tend to increase the corrosion locally. This has not been noticed on the designated test masts but at regular masts close to station areas with public access. This knowledge should be added to any future design philosophy if masts of weathering steel are located in areas where graffiti is likely to be a problem. At some point during the years a reflective self-adhesive tape was introduced at the bottom of many masts. But as it kept the corrosive layer damp it lead to large increase in local corrosion. Thus the tape is now re-fixed with spacers similar to the clamps and this method seems to work very well.



Fig.6. a) Dismantled clamps with and without spacers; b) CHS where both clamps have been mounted; c) CHS where clamp with spacer was located; d) CHS where clamp without spacer was located

6 PROPOSED CORROSION RATES FOR DESIGN

Many various equations for predicting the atmospheric corrosion of weathering steels are given in the literature. These vary in type from power models, numerical curve fits, power-linear models, bi-linear models etc. Most experimental data is found to follow the relationship $C = A \cdot t^n$ [5] and this relationship is also what has been used in design of the structures for the Banedanmark. The benefit of this type of equation is, that short term test-data can be used to predict long term exposure. But it is not as simple a design rule as a linear equation would be.

It is only in very rare occasions that structures of civil engineering are designed with an expected service life below 10 years, and thus the corrosion rate of weathering steel before the rust layer has stabilized is not of great importance for these structures. Once the dense layer is formed, it would be fair to consider a constant rate of corrosion. The design of civil structures in recent years turns towards requirements of very long term service life i.e. 100 years or more. This kind of requirement adds uncertainty to the design, as it is very difficult to predict the behaviour, when experience with such long time of exposure does not exist. Thus a linear curve might add a bit of conservatism compared to the power curve on the very long term, but that might be fair as a rural environment might have changed to something else during the 100 years.

Based on the resumption of the test-regime in 2016, where exposure time up to 28 years was considered at one location, it appears that mast structures for supporting electrical overhead line equipment in the non-coastal areas of Denmark would be safe to design considering atmospheric corrosion of

$$C = YCL \cdot t + ICL \quad (1)$$

where t is the time of exposure in years
 YCL is the yearly corrosion loss [$\mu\text{m}/\text{year}$] – typical 6 $\mu\text{m}/\text{year}$
 ICL is the initial corrosion loss [μm] – typical 100 μm
 C is the corrosion in μm

YCL and ICL will depend of the corrosion category and the orientation of the exposed surface.

7 FUTURE WORK

As only one of the test-masts was considered after long time exposure, it is necessary to measure the corrosion on further test-masts and gather further information before a solid conclusion on the expected remaining service life can be made. This would include structures exposed to marine environment of the North Sea. Since the most of the masts are still standing and the thickness of the plates was measured from the start, we are in a unique position to gain more knowledge about the corrosion of weathering steel after long term exposure.

This work is planned to commence during 2017.

8 CONCLUSIONS

In general the masts can be divided into coastal and medium environment. It is assumed that coastal environment approximate covers any mast placed closer than 1000 m from any coast in Denmark.

It has been shown that the design service life of 50 years for a sacrificial thickness of 0.6 mm is conservative. With the mentioned typical values the service life can be increased to 85 years, but it would need to be studied further at other locations too.

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